CREDIT CARD FRAUD PATTERN & LOSS ANALYSIS

Management Recommendation Memo | FY2025

Decision: Deploy R001 as the primary manual-review loss-prevention control

Executive recommendation. Deploy R001: E_COMMERCE + (Electronics or Gift Cards) + transaction amount >= ■20,000. On the validation period, R001 generated 616 alerts, achieved 70.13% precision, captured ■18.55M of fraud value, and produced ■14.73M of expected net savings after review and customer-friction costs. The rule uses approximately 30.8% of the 500-alert/month review capacity.

1. Business problem

The bank needs to reduce credit-card fraud losses while avoiding excessive customer friction and an unmanageable manual-review workload. The analysis focuses on loss concentration, fraud patterns, rule thresholds, validation performance, and financial value.

2. Executive baseline

Metric	FY2025 result
Total transaction value	■3.76B
Confirmed fraud transactions	10,488
Fraud rate	0.70%
Gross chargeback loss	■95.29M
Recovered amount	■26.51M
Net chargeback loss	■68.78M

The transaction-level fraud baseline uses confirmed chargebacks linked to the cleaned transaction fact. The overall chargeback loss baseline includes all 10,500 chargebacks, including 12 retained orphan chargeback records.

3. Key analytical findings

E-commerce concentration: E-commerce accounts for ■57.75M of net chargeback loss, approximately 84% of total net loss.

Category concentration: Electronics and Gift Cards together account for approximately ■51.78M, about 75% of net chargeback loss.

High-value concentration: Transactions above ■25K represent only about 0.32% of transaction volume but account for about 55.87% of fraud value.

Country mismatch: Country mismatch has a 1.22% fraud rate versus 0.60% for matched transactions, making it a useful supporting signal rather than a standalone rule.

Card age: Fraud rates remain broadly flat across card-age cohorts, so card age is not a strong standalone trigger.

Velocity: Simple 10-minute and 24-hour velocity counts show limited standalone separation and are better used in combination with other signals.

4. Recommended rule — R001

Rule logic: IF channel = E_COMMERCE AND category IN {Electronics, Gift Cards} AND transaction amount >= ■20,000 THEN generate a manual-review alert.

Why ■20,000? Threshold analysis showed the expected trade-off: lower thresholds capture more fraud but generate more alerts and lower precision; higher thresholds improve precision but sacrifice fraud value. ■20K was selected as the operating point for the final rule.

5. Validation performance and economics

Metric	R001	R002
Validation alerts	616	781
Precision	70.13%	91.68%
Recall	12.26%	20.32%
Fraud value caught	■18.55M	■18.49K
Expected net savings	■14.73M	-■108.85K

R001 is preferred even though R002 has higher precision. R001 captures materially more monetary loss and produces positive expected economics. R002 is better positioned as a targeted preventive card-testing control.

6. R001 business case

Business-case component	Validation result
Fraud value caught	■18.55M

Business-case component	Validation result
Assumed fraud prevention rate	80%
Expected prevented loss	■14.84M
Review cost	■92.40K
Customer-friction cost	■18.40K
Expected net savings	■14.73M
Average alerts/month	154
Review capacity	500 alerts/month
Capacity utilization	30.8%

7. Typology coverage and control limitations

Fraud typology	Validation fraud events	R001 recall
Account takeover	693	61.90%
Card testing	716	0.00%
CNP high-risk merchant	622	0.00%
Compromised terminal	506	0.20%
Cross-border spike	509	0.20%
Impossible travel	478	0.21%

R001 is primarily an account-takeover/high-value e-commerce loss-prevention control. It should not be treated as a complete fraud-detection system; complementary controls are required for typologies not covered by R001.

8. Deployment recommendation

• Deploy R001 as the primary manual-review control in a controlled production/pilot phase.

• Retain R002 as a targeted preventive/card-testing control rather than a manual-review rule under the current cost assumptions.

• Monitor alert volume, precision, prevented fraud value, customer friction, and reviewer capacity after deployment.

• Recalibrate thresholds using fresh labeled outcomes and time-based backtesting as the fraud environment changes.

• Maintain complementary controls for fraud typologies not covered by R001.

9. Assumptions and limitations

The business case assumes an 80% fraud-prevention rate, ■150 review cost per alert, ■100 customer-friction cost per legitimate alert, and a 500-alert/month review capacity. These are scenario assumptions rather than observed production costs. The dataset is synthetic and includes injected fraud typologies and data-quality issues designed for analytical testing. Validation is time-based, with development and validation periods separated to reduce future leakage. Results should therefore be treated as a project recommendation framework and recalibrated with real bank outcomes before production deployment.